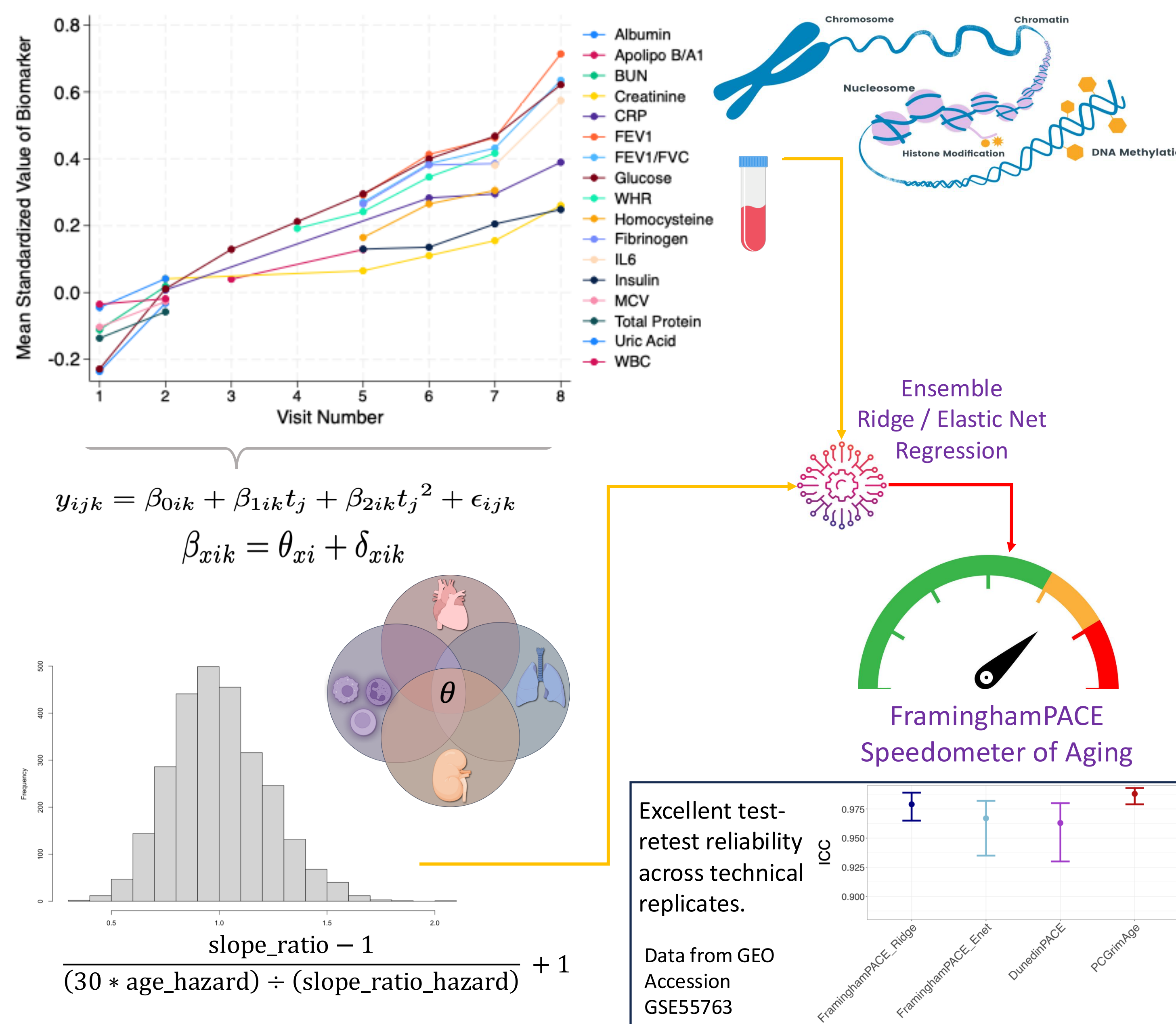


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Background

BACKGROUND. The leading algorithm-based biomarkers of aging estimate Biological Age, the age at which a person's biology (or surrogates like mortality risk) would appear normal in some reference population (e.g. Horvath 2013, Levine et al. 2018, Lu et al. 2019). These biomarkers function like odometers for the aging process, marking how much aging a person has experienced up to the time their DNA is sampled. An alternative approach is to estimate Pace of Aging, the rate of decline in system integrity (Belsky et al. 2015, 2022; Miller et al. 2023). Pace of Aging biomarkers function like speedometers, capturing the rate at which a person's biology is changing due to aging. In the CALERIE trial, which evaluated the effect of 2y of calorie restriction in healthy, non-obese adults, the epigenetic speedometer DunedinPACE was responsive to intervention, whereas the leading odometer epigenetic clocks, including GrimAge, were not. We hypothesized the greater sensitivity of DunedinPACE owed to its nature as a speedometer. However, several other explanations have been proposed for the CALERIE result, including the demographic similarity between the relatively young cohort in which DunedinPACE was developed and the trial, whose participants were young-midlife adults. To test our hypothesis, we created a new epigenetic speedometer for Pace of Aging from the same data used to develop GrimAge.

Training

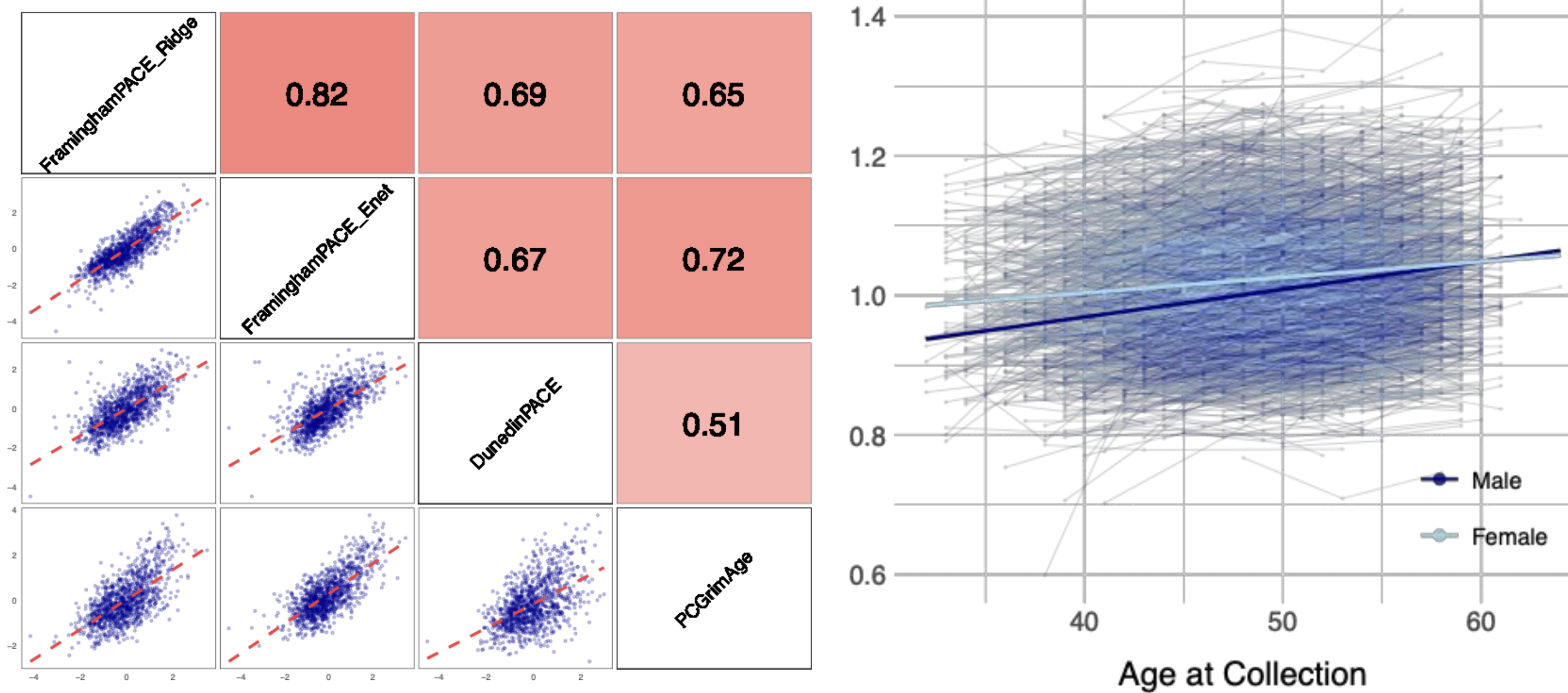


METHOD. We developed FraminghamPACE from longitudinal analysis of change in 17 biomarkers of organ system functioning over 30y of follow-up in the Framingham Heart Study Offspring Cohort. We used a hierarchical model to estimate a shared slope across biomarkers for each participant (θ). We next normalized θ to reflect Pace of Aging, defined as years of biomarker change per calendar year. This involved computing the ratio of each participant's θ to the cohort mean and then scaling it according to the mortality hazard per year of age. Finally, we modeled the normalized θ from DNA methylation data collected at the end of follow-up using ensemble Ridge and elastic-net regression models with age and sex included as unpenalized regressors. Ensemble models were employed to improve model fit, stability, and generalizability by aggregating predictions across multiple training subsamples. We included as features in our analysis CpG sites identified as being measured consistently across iterations of Illumina arrays. We projected the resulting algorithm onto data from epidemiologic cohorts to establish correlations with age and other estimates of biological aging and to validate FraminghamPACE as a predictor of healthspan and lifespan. Finally, we projected FraminghamPACE on CALERIE trial data and repeated our original analysis.

Correlation with Age and Epigenetic Clocks



FraminghamPACE is correlated with DunedinPACE and GrimAge and indicates accelerating aging with advancing chronological age.



CARDIA enrolled White and Black men and women aged 18-30y. DNA methylation data were generated from samples collected at follow-ups 15, 20, 25, and 30 years after baseline. Our analysis included EPIC array data for n=2,023 individuals with up to four repeated measures (6,214 observations). Correlation matrix shows data from DNAm baseline at y15.

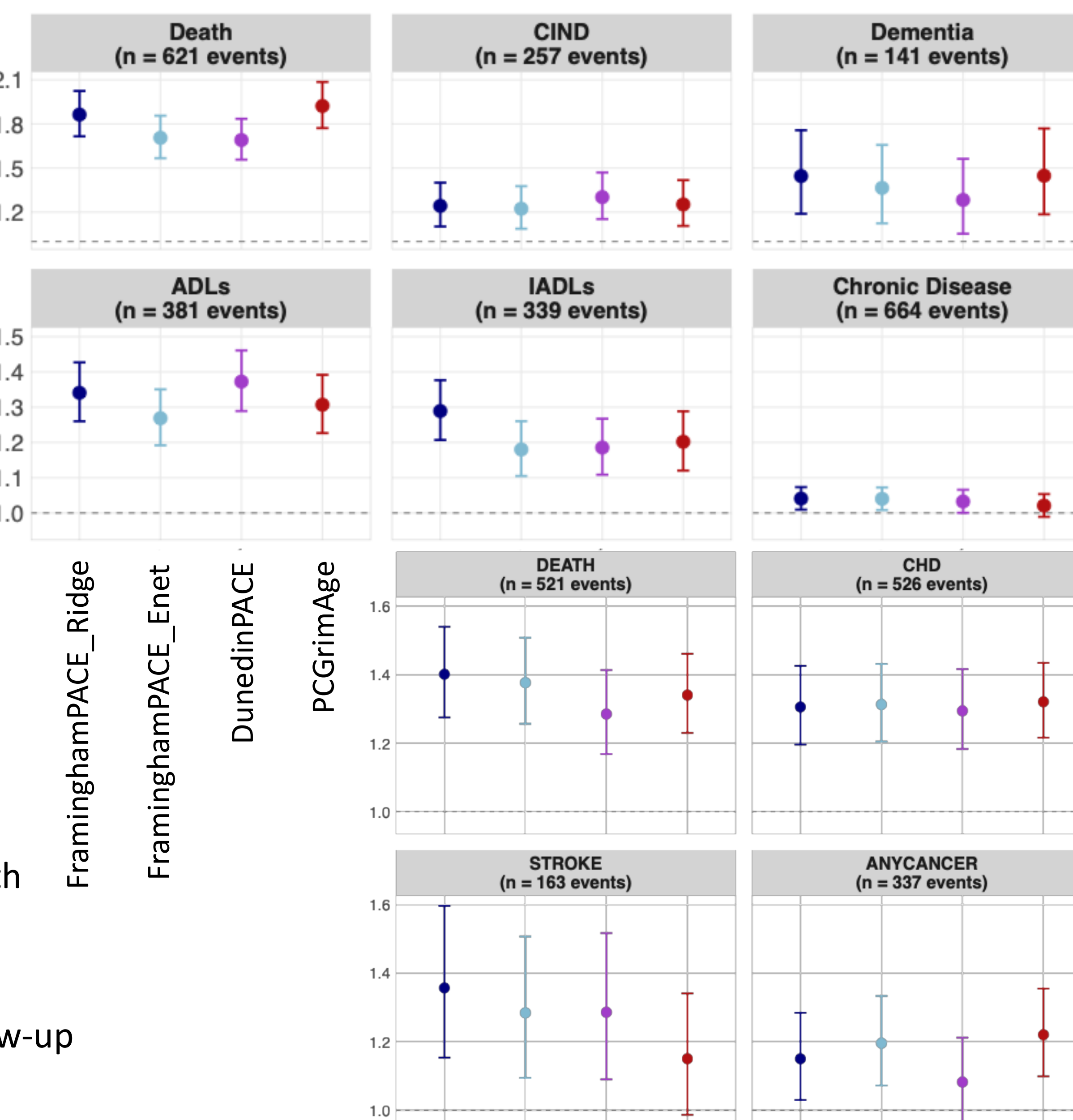
Analysis of Healthspan and Lifespan



US Health &
Retirement
Study
n=3,433
Up to 6y
follow-up



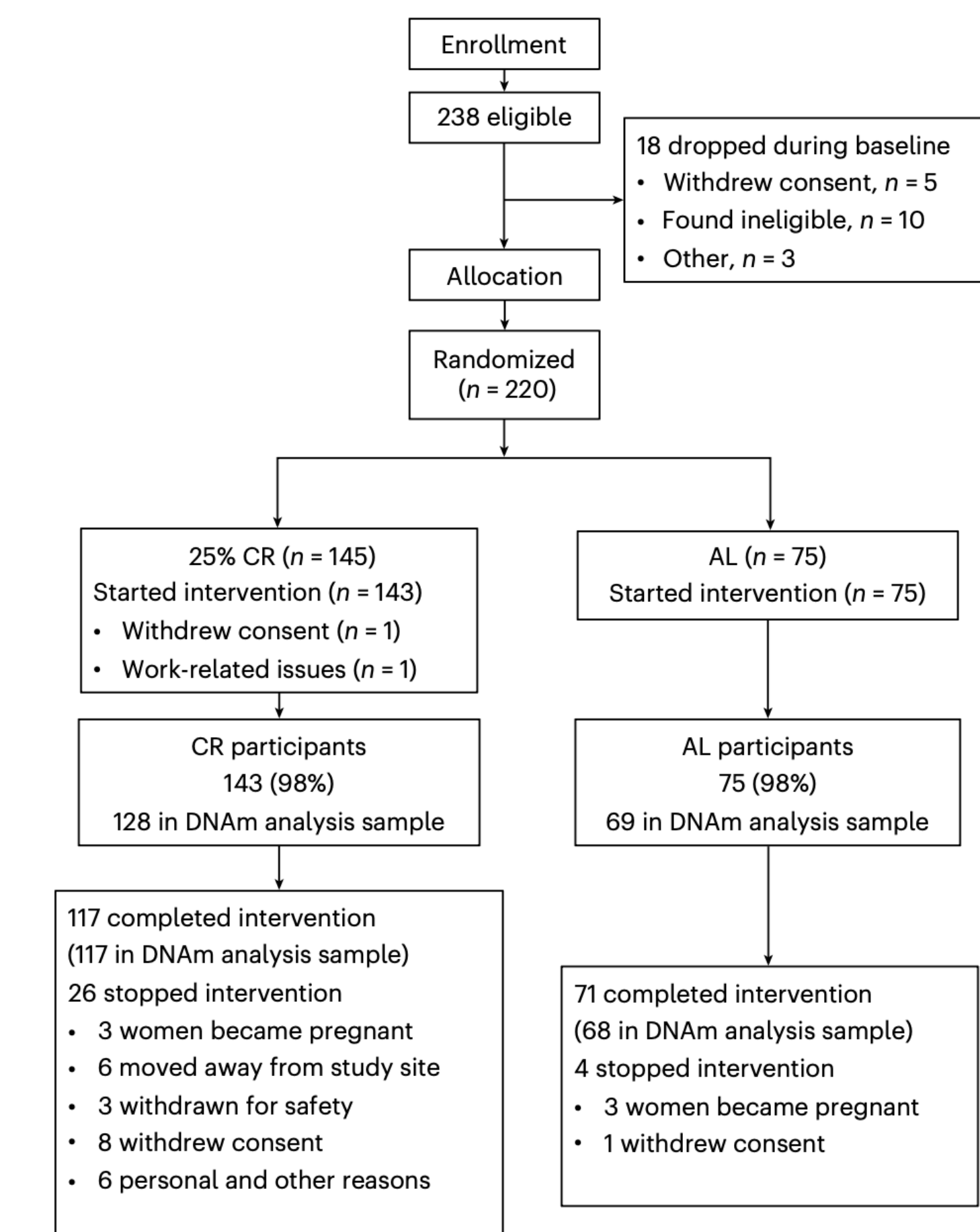
Women's Health Initiative
n=1,835
Up to 21y follow-up



CALERIE Trial Analysis

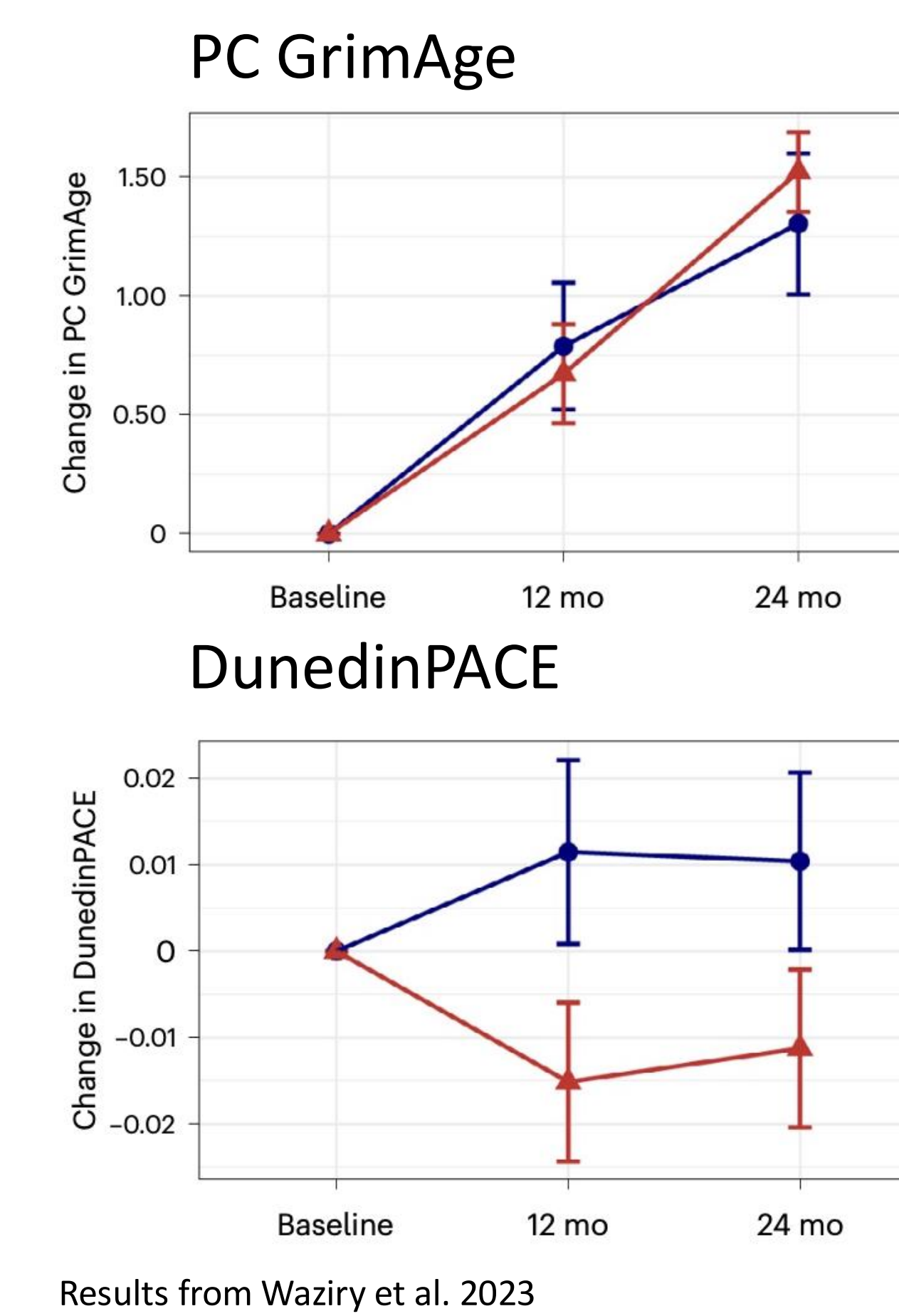


Comprehensive Assessment of Long-term Effects of Reducing Intake of Energy (CALERIE) Phase 2 was a multi-center, randomized controlled trial conducted at three clinical centers in the United States. It aimed to evaluate the time-course effects of 25% CR (that is, intake 25% below the individual's baseline level) over a 2-yr period in healthy adults (men aged 21–50 yr, premenopausal women aged 21–47 yr) with body mass index (BMI) in the normal weight or slightly overweight range (BMI 22.0–27.9 kg m²). Participants were randomly assigned at a ratio of 2:1 to a CR behavioral intervention or to an ad libitum (AL) control group stratified by site, sex and BMI. Of 238 eligible individuals, CALERIE randomized N = 220 participants (145 CR intervention and 75 AL control). Participants in the CR group were prescribed a 25% restriction in calorie intake based on energy requirements estimated from two 2-week doubly labeled water (DLW) measurement periods at baseline. The precise level of CR achieved was quantified by comparing energy intake (determined periodically throughout the trial by the DLW method) during the CR intervention with baseline energy intake.

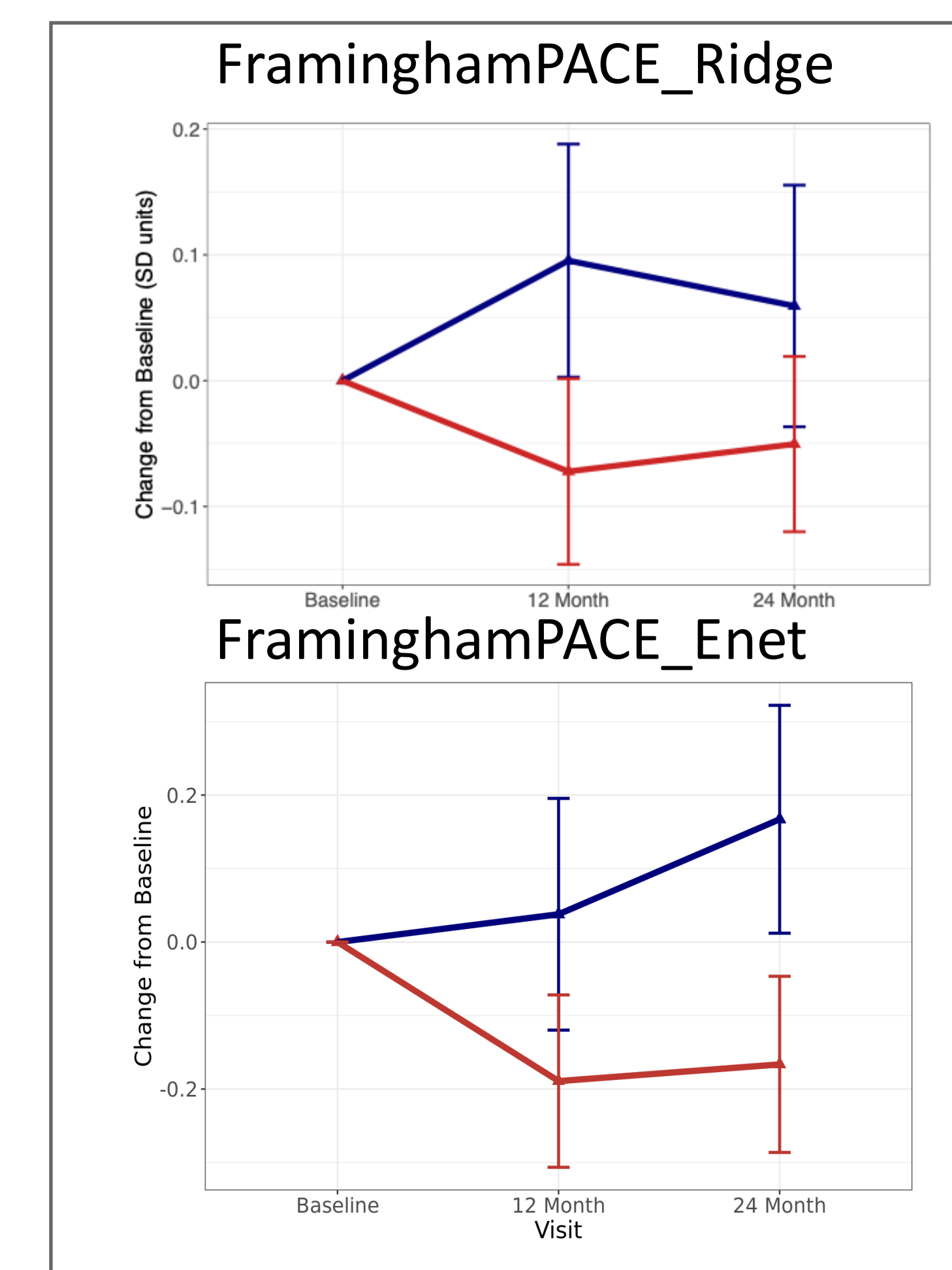


Treatment group

AL CR



Results from Waziry et al. 2023



Discussion

DISCUSSION. FraminghamPACE is a new epigenetic speedometer of the Pace of Aging. It has similar properties to DunedinPACE, including comparable prediction of morbidity and mortality. Critically, FraminghamPACE shows parallel response to CALERIE intervention to DunedinPACE. This stands in contrast to the results for GrimAge, which was developed in the same cohort. This result supports the hypothesis that Pace of Aging offers advantages as a label in ML/AI analysis of omics data to derive sensitive endpoints for clinical trials of geroprotective intervention.

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